

4.5 Fact 5: Labor market adjustments are visible in employment more than compensation

In addition to employment we observe workers’ annual base compensation. We use this information to test for labor market adjustment along the compensation margin.²⁰ Salary data are deflated to 2017 dollars using the PCE index.²¹

Results for various occupations are in Figure A11. The findings indicate less divergence in compensation compared to employment across more and less exposed occupations. Figure 5 shows results by age and Eloundou et al. (2024)-based exposure quintile. We find little difference in compensation trends by age or exposure quintile.

Prior work by Autor and Thompson (2025) notes that technology that replaces inexpert tasks may reduce occupational employment but increase occupational wages; technology that replaces expert tasks may do the opposite. The sign of the wage effect depends on the overall share of tasks displaced as well as the whether these tasks are expert or inexpert. The limited changes we find for wages suggest that these effects may be offsetting, at least in the short run. Alternatively, the results could be explained by wage stickiness in the short run, consistent with recent evidence from Davis and Krolikowski (2025).

4.6 Fact 6: Findings are largely consistent under alternative sample constructions

We test the robustness of these results to alternative sample constructions and robustness checks.

Excluding Technology Occupations One possibility is that our results are explained by a general slowdown in technology hiring from 2022 to 2023 as firms recovered from the COVID-19 Pandemic.²² Figure A12 shows employment changes by age and exposure quintile after excluding computer occupations, corresponding to 2010 SOC codes that start with 15-1. Figure A13 shows results when excluding firms in information technology or computer systems design (NAICS codes

²⁰Total compensation may additionally include bonuses, overtime pay, commissions, equity, tips, and other items. These may have a greater impact on overall compensation in certain professions and age groups than others.

²¹In contrast to the series for employment, results for compensation end in August 2025, the most recently available month for the PCE index.

²²Under the Tax Cuts and Jobs Act, amendments to Internal Revenue Code §174 enacted in 2022 also disallowed companies from immediately deducting R&D expenditures, including software development costs.

51 and 5415). Results are quite similar, consistent with the above case studies showing employment changes across various occupations. Results with firm-time fixed effects in Figure 4 further show that our findings are robust to firm- or industry-level shocks that impact general hiring trends. These results indicate that our findings are not specific to technology roles.

Remote Work Figures A14 and A15 show results for occupations amenable to remote work (telework) and those that are not, according to Dingel and Neiman (2020).²³ We find that, for young workers, more exposed occupations have slower employment growth, both in teleworkable occupations and in non-teleworkable occupations. The results for non-teleworkable occupations in particular suggest that our findings are not driven by outsourcing or work-from-home disruptions, at least solely.²⁴

Longer Sample Figure A16 shows results when extending the balanced sample of firms to 2018. This reduces the sample size and makes the data somewhat noisier. Nonetheless, the trends remain largely ordered by exposure in the post-GPT era, whereas this is not the case before 2022. A concern is that for the Eloundou et al. (2024) measures the most exposed quintile had slower employment growth starting around 2020. This is not the case for the Anthropic exposure measures, shown in Figures A17, A18, and A19. For these measures the most exposed groups have comparable employment growth throughout the period before generative AI, with divergent trends afterwards.

Changes in Education Another possibility is that our results stem from worsening education outcomes during the COVID-19 Pandemic, as more educated workers have greater average AI exposure (Kuhfeld and Lewis, 2025).²⁵ In Figure A20 we show trends for occupations in which greater than 70% of workers have a college degree according to the 2017 American Community Survey (ACS).²⁶ In Figure A21 we show trends for occupations in which fewer than 30% of workers

²³Whether an occupation is amenable to remote work is positively correlated with AI exposure. Only two teleworkable occupations fall in the lowest quintile of estimated AI exposure according to the GPT-4 β measure. Likewise few non-teleworkable occupations fall in the highest quintile of AI exposure. For this reason in Figure A14 we pool together the two lowest AI exposure quintiles into one group. In Figure A15 we pool together the two highest AI exposure quintiles.

²⁴Non-teleworkable occupations with high AI exposure include bank tellers, travel agents, and tax preparers.

²⁵Chandar (2025a) finds that declines in average skill levels for college graduates explain a sizable share of the slowdown in the growth of the college wage gap in recent decades.

²⁶Not a single occupation in the first quintile of GPT-4 β based exposure measure has a college share above 35%, so that quintile is excluded from the results in Figure A20.

have a college degree.

Occupations with a high share of college graduates have declining employment overall, with muted differences between more-exposed and less-exposed occupations compared to our main results. In contrast, occupations with a low share of college graduates have rising overall employment, with the least AI-exposed occupations growing and the most exposed occupations declining in employment. Further, for lower college share occupations, the dispersion in employment outcomes is visible in higher age groups as well, with workers up to age 40 showing separation in employment trends by AI exposure. These findings suggest that deteriorating education outcomes cannot fully explain our main results. They also suggest that for non-college workers, experience may serve as less of a buffer to labor market disruption than for college workers.

Occupational Interest Rate Exposure While estimates with firm-time effects control for overall hiring trends within firms, another possibility is that even within firms occupations with high AI exposure have high interest rate exposure, leading to contraction in hiring. We use occupational interest rate exposure data from [Zens et al. \(2020\)](#) to test for the correlation between interest rate exposure and AI exposure. Figure [A22](#) in fact finds that AI exposure and interest rate exposure are negatively correlated overall, with occupations such as construction having high interest rate exposure and low AI exposure. Figures [A23](#) and [A24](#) repeat our analysis separately for occupations below and above the median in interest rate exposure. Both cases are consistent with our main results, suggesting that occupational differences in interest rate exposure do not drive our results.

Other Robustness Checks Figures [A25](#) and [A26](#) show results separately for men and women. The results are similar, suggesting that diverging prospects for men and women are not driving our findings. Figure [A27](#) shows that results are similar when we do not take a balanced sample of firms. Figure [A28](#) shows similar results when including part-time and temporary workers.²⁷

Comparison to CPS Data A useful benchmark for our findings is to compare them to estimates from the monthly Current Population Survey (CPS). In Online Appendix [C](#) we perform similar analyses in the CPS, finding high levels of noise consistent with small sample sizes in fine age-

²⁷Another possibility is that employment trends reflect Covid-19 stimulus checks distorting labor supply. However, these payments were income-conditioned, and AI-exposed occupations average higher incomes ([Kochhar, 2023](#)), making this channel unlikely in explaining our findings.

occupation cells. Future work should study employment trends in other large-scale data sources such as the American Community Survey (ACS) or data from Revelio Labs as in [Hosseini and Lichtinger \(2025\)](#) and [Klein Teeselink \(2025\)](#).

5 Conclusion

We document six facts about the recent labor market effects of artificial intelligence.

- We find substantial employment declines for early-career workers in occupations most exposed to AI, such as software development and customer support.
- While economy-wide employment continues to grow, employment growth for young workers has been stagnant.
- Entry-level employment has declined in applications of AI that *automate* work, with muted effects for those that *augment* it.
- After conditioning on firm-time effects, young workers experienced a 16% relative employment decline in the most exposed occupations.
- Labor market adjustments are more visible in employment than in compensation.
- These patterns hold across various alternative analyses.

While our main estimates may be influenced by factors other than generative AI, our results are consistent with the hypothesis that generative AI has begun to affect entry-level employment significantly.

The adoption of new technologies typically leads to heterogeneous effects across workers, resulting in adjustment periods as workers reallocate from displaced forms of work to new forms with growing labor demand ([Autor et al., 2024](#)). Past transitions such as the IT revolution ultimately led to robust growth in employment and real wages following physical and human capital adjustments, with some workers benefiting more than others ([Bresnahan et al., 2002](#); [Brynjolfsson et al., 2021](#)).

Tracking employment trends on an ongoing basis will help determine if the adjustment to AI follows a similar pattern. We will continue monitoring these outcomes to assess whether the trends